

5th Place Solution — DINOv3 Semantic Segmentation

Rank	5
Team	Guanshuo Xu
Author	Guanshuo Xu
Topic ID	697699
Version	Original

Source: <https://www.kaggle.com/competitions/recodai-luc-scientific-image-forgery-detection/discussion/697699>
Retrieved with Kaggle CLI on 2026-07-18.

I would like to thank Kaggle and Recod.ai/LUC for hosting this interesting competition. I finished 5th on the private leaderboard.

Final scores:

- Public LB: **0.422**
- Private LB: **0.376**
- [inference code](#)
- **training code attached at the end**

Overview

My solution focused on simplicity and robustness. I treated the task primarily as a semantic segmentation problem.

For training mask generation, when an image contained multiple copy-move instances, I used the union of all manipulated regions as the training target. In other words, the model was trained as a binary manipulated-region segmenter rather than as an instance segmentation model.

At inference time, I also did not separate multiple copy-move instances. This was a deliberate simplification.

Two factors contributed most significantly to the final score:

1. Using external datasets to improve generalization.
2. Using the strongest available pretrained DINOv3 backbone.

Validation

I mainly validated my models on the 48 supplemental images provided by the host, because these multi-panel images were the closest match to the actual test set.

Backbone and Segmentation Head

I used pretrained DINOv2/DINOv3 backbones with a very simple segmentation head.

The model takes the final patch tokens from the ViT backbone, reshapes them into a 2D feature grid, and applies a single 1×1 Conv2d head to produce one logit map.

During training, I downsampled the ground-truth mask to match the patch-grid resolution:

- 14×14 for DINOv2
- 16×16 for DINOv3

During inference, after predicting the low-resolution mask, I simply resized it back to the original image size using bilinear interpolation.

Augmentations

The most useful augmentations were simple image-level augmentations:

```
A.RandomRotate90(p=1.0)
A.HorizontalFlip(p=0.5)
A.GaussianBlur(p=0.15)
A.ToGray(...)
A.HueSaturationValue(...)
A.RandomBrightnessContrast(p=0.9)
A.ImageCompression(p=0.25)
```

I used fairly aggressive color augmentation because the manipulation signal is not always color-specific. The model needed to learn structural and local consistency cues rather than memorize color distributions.

Important Training Choices

- BF16 autocast
- Full fine-tuning
- FlashAttention-2
- Input image size: 1024×1024
- Vanilla BCE loss

Important Inference Choices

- 8-bit inference to fit within the 16GB memory limit of a T4 GPU
- SDPA attention
- Input image size: 1024×1024
- Four-way `rotate90` test-time augmentation

Post-processing

The post-processing pipeline was simple:

1. Create a high-confidence binary mask using a fixed threshold.
2. Compute the area of this high-confidence mask by summing its active pixels.
3. If the high-confidence area is smaller than `min_area`, classify the sample as authentic and stop.
4. Create the final binary mask using the configured threshold `mask_thr`.
5. If the final mask contains no active pixels, classify the sample as authentic and stop.

Early Results

I started with a DINOv2 model. Its validation score on the supplemental images was only **0.092**, although the validation score on a split of the provided training set was much higher.

After inspecting the training data, I found that although the dataset was not small, it contained only a limited variety of image types. As a result, the fine-tuned model did not generalize well to the supplemental set, which was visually quite different from the training set.

External Datasets

To improve generalization, I searched for external datasets that could increase the diversity of the training data. I eventually added the following datasets:

- [CASIA](#)
- [GRIP](#)
- [FAU](#)

Validation results on the supplemental images were:

Training data	Validation score
Competition training set only	0.092
Competition training set + CASIA	0.185
Competition training set + GRIP	0.095
Competition training set + FAU	0.129
Competition training set + CASIA + GRIP + FAU	0.240

Although these external datasets are not biomedical, their diversity helped the model learn more general copy-move detection cues.

Better Backbones

Because copy-move detection is a difficult task, stronger pretrained backbones made a large difference.

Upgrading the backbone from **DINOv2 Giant** to **DINOv3 Huge+** improved the validation score from the earlier range to around **0.35**. Since these two models are relatively close in size, I believe the improvement mainly came from the stronger pretraining of DINOv3.

I then used the original **DINOv3 7B** model, which further improved the validation score to **0.51**. With additional hyperparameter tuning, the best validation score exceeded **0.56**.

Overall, the final solution remained simple: a strong pretrained ViT backbone, a minimal segmentation head, diverse training data, and lightweight post-processing.